



Module II

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2. Linear System of Algebraic Equations

2.1 Introduction

Consider a system of n linear algebraic equations in n unknowns

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2$$

$$\cdots \quad \cdots \quad \cdots \quad \cdots$$

$$a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n = b_n$$

where a_{ij} , $i = 1, 2, \dots, n$, are the known coefficients, b_i , $i = 1, 2, \dots, n$, are the known right hand side values and x_i , $i = 1, 2, \dots, n$ are the unknowns to be determined.

In matrix notation we write the system as

$$\mathbf{Ax} = \mathbf{b} \tag{2.1}$$

$$\text{where } \mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \cdots \\ x_n \end{bmatrix}, \text{ and } \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \cdots \\ b_n \end{bmatrix}.$$

The matrix $[\mathbf{A} \mid \mathbf{b}]$, obtained by appending the column $\mathbf{b}$ to the matrix $\mathbf{A}$ is called the *augmented matrix*. That is

$$[\mathbf{A} \mid \mathbf{b}] = \left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & b_n \end{array} \right]$$

We define the following.

- i) The system of equations (1) is consistent (has at least one solution), if

$$\text{rank}(\mathbf{A}) = \text{rank}[\mathbf{A} \mid \mathbf{b}] = r.$$

If $r = n$, then the system has unique solution.

If $r < n$, then the system has $(n - r)$ parameter family of infinite number of solutions.

- ii) The system of equations (1) is inconsistent (has no solution) if

$$\text{rank}(\mathbf{A}) \neq \text{rank}[\mathbf{A} \mid \mathbf{b}].$$

We assume that the given system is consistent.

The methods of solution of the linear algebraic system of equations (2.1) may be classified as direct and iterative methods.

- i) *Direct methods* produce the exact solution after a finite number of steps (disregarding the round-off errors). In these methods, we can determine the total number of operations (additions, subtractions, divisions and multiplications). This number is called the operational count of the method.
- ii) *Iterative methods* are based on the idea of successive approximations. We start with an initial approximation to the solution vector $\mathbf{x} = \mathbf{x}_0$, and obtain a sequence of approximate vectors $\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_k, \dots$, which in the limit as $k \rightarrow \infty$, converge to the exact solution vector $\mathbf{x}$.
Now, we derive some direct methods.

2.2 Direct Methods

If the system of equations has some special forms, then the solution is obtained directly.

We consider two such special forms.

- a) Let $\mathbf{A}$ be a diagonal matrix, $\mathbf{A} = \mathbf{D}$. That is, we consider the system of equations $\mathbf{D}\mathbf{x} = \mathbf{b}$ as

$$\begin{array}{ccccccc} a_{11}x_1 & & & & & & = b_1 \\ & a_{22}x_2 & & & & & = b_2 \\ & & \dots & \dots & \dots & & \dots \\ & & & a_{n-1,n-1}x_{n-1} & & & = b_{n-1} \\ & & & & a_{nn}x_n & & = b_n \end{array} \quad (2.2)$$

This system is called a *diagonal system of equations*. Solving directly, we obtain

$$x_i = \frac{b_i}{a_{ii}}, a_{ii} \neq 0, i = 1, 2, \dots, n. \quad (2.3)$$

- b) Let $\mathbf{A}$ be an upper triangular matrix, $\mathbf{A} = \mathbf{U}$. That is, we consider the system of equations $\mathbf{U}\mathbf{x} = \mathbf{b}$ as

$$\begin{array}{ccccccc} a_{11}x_1 & + & & & & & = b_1 \\ & a_{22}x_2 & & & & & = b_2 \\ & & \dots & \dots & \dots & & \dots \\ & & & a_{n-1,n-1}x_{n-1} + a_{n-1,n}x_n & & & = b_{n-1} \\ & & & & a_{nn}x_n & & = b_n \end{array} \quad (2.4)$$

This system is called an *upper triangular system of equations*. Solving for the unknowns in

the order $x_n, x_{n-1}, \dots, x_1$, we get

$$\begin{aligned} x_n &= b_n / a_{nn'} \\ x_{n-1} &= (b_{n-1} - a_{n-1,n}x_n) / a_{n-1,n-1'} \\ &\dots \dots \dots \dots \\ x_1 &= \frac{\left(b_1 - \sum_{j=2}^n a_{1,j}x_j \right)}{a_{11}} = \left(b_1 - \sum_{j=2}^n a_{1,j}x_j \right) / a_{11} \end{aligned} \quad (2.5)$$

The unknowns are obtained by back substitution and this procedure is called the *back substitution* method.

Therefore, when the given system of equations is one of the above two forms, the solution is obtained directly.

Before we derive some direct methods, we define elementary row operations that can be performed on the rows of a matrix.

Elementary row transformations (operations) The following operations on the rows of a matrix **A** are called the *elementary row transformations (operations)*.

- Interchange of any two rows. If we interchange the i th row with the j th row, then we usually denote the operation as $R_i \leftrightarrow R_j$.
- Division/multiplication of any row by a non-zero number p . If the i th row is multiplied by p , then we usually denote this operation as pR_i .
- Adding/subtracting a scalar multiple of any row to any other row. If all the elements of the j th are multiplied by a scalar p and added to the corresponding elements of the i th row, then, we usually denote this operation as $R_i \leftarrow R_i + pR_j$. Note the order in which the operation $R_i + pR_j$ is written. The elements of the j th row remain unchanged and the elements of the i th row get changed.

These row operations change the form of **A**, but do not change the row-rank of **A**. The matrix **B** obtained after the elementary row operations is said to be row equivalent with **A**. In the context of the solution of the system of algebraic equations, the solution of the new system is identical with the solution of the original system.

The above elementary operations performed on the columns of **A** (column C in place of row R) are called *elementary column transformations (operations)*. However, we shall be using only the elementary row operations.

In this section, we derive two direct methods for the solution of the given system of equations, namely, Gauss elimination method and Gauss-Jordan method.

2.3 Gauss Elimination Method

The method is based on the idea of reducing the given system of equations $\mathbf{Ax} = \mathbf{b}$, to an upper triangular system of equations $\mathbf{Ux} = \mathbf{z}$, using elementary row operations. We know that these two systems are equivalent. That is, the solutions of both the systems are identical. This reduced system $\mathbf{Ux} = \mathbf{z}$, is then solved by the back substitution method to obtain the solution vector $\mathbf{x}$.

We illustrate the method using the 3×3 system

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3 \end{aligned} \quad (2.6)$$

We write the augmented matrix $[\mathbf{A} \mid \mathbf{b}]$ and reduce it to the following form

$$[\mathbf{A} \mid \mathbf{b}] \xrightarrow{\text{Gauss elimination}} [\mathbf{U} \mid \mathbf{z}]$$

The augmented matrix of the system (2.6) is

$$\left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{array} \right] \quad (2.7)$$

First Stage of elimination

We assume $a_{11} \neq 0$. This element a_{11} in the 1×1 position is called the *first pivot*. We use this pivot to reduce all the elements below this pivot in the first column as zeros. Multiply the first row in (7) by a_{21}/a_{11} and a_{31}/a_{11} respectively and subtract from the second and third rows. That is, we are performing the elementary row operations $R_2 - (a_{21}/a_{11})R_1$ and $R_3 - (a_{31}/a_{11})R_1$ respectively. We obtain the new augmented matrix as

$$\left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ 0 & a_{22}^{(1)} & a_{23}^{(1)} & b_2^{(1)} \\ 0 & a_{32}^{(1)} & a_{33}^{(1)} & b_3^{(1)} \end{array} \right] \quad (2.8)$$

$$\begin{aligned} \text{where} \quad a_{22}^{(1)} &= a_{22} - \left(\frac{a_{21}}{a_{11}}\right)a_{12}, a_{23}^{(1)} = a_{23} - \left(\frac{a_{21}}{a_{11}}\right)a_{13}, b_2^{(1)} = b_2 - \left(\frac{a_{21}}{a_{11}}\right)b_1, \\ a_{32}^{(1)} &= a_{32} - \left(\frac{a_{31}}{a_{11}}\right)a_{12}, a_{33}^{(1)} = a_{33} - \left(\frac{a_{31}}{a_{11}}\right)a_{13}, b_3^{(1)} = b_3 - \left(\frac{a_{31}}{a_{11}}\right)b_1. \end{aligned}$$

Second stage of elimination

We assume $a_{22}^{(1)} \neq 0$. This element $a_{22}^{(1)}$ in the 2×2 position is called the *second pivot*. We use this pivot to reduce the element below this pivot in the second column as zero. Multiply the second row in (8) by $a_{32}^{(1)}/a_{22}^{(1)}$ and subtract from the third row. That is, we are performing the elementary row operation $R_3 - a_{32}^{(1)}/a_{22}^{(1)}R_2$. We obtain the new augmented matrix as

$$\left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ 0 & a_{22}^{(1)} & a_{23}^{(1)} & b_2^{(1)} \\ 0 & 0 & a_{33}^{(2)} & b_3^{(2)} \end{array} \right] \quad (2.9)$$

$$\text{where} \quad a_{33}^{(2)} = a_{33}^{(1)} - \left(\frac{a_{32}^{(1)}}{a_{22}^{(1)}}\right)a_{23}^{(1)}, b_3^{(2)} = b_3^{(1)} - \left(\frac{a_{32}^{(1)}}{a_{22}^{(1)}}\right)b_2^{(1)}.$$

The element $a_{33}^{(2)} \neq 0$ is called the *third pivot*. This system is in the required upper triangular form $[\mathbf{U} \mid \mathbf{z}]$. The solution vector $\mathbf{x}$ is now obtained by back substitution.

$$\text{From the third row, we get} \quad x_3 = b_3^{(2)}/a_{33}^{(2)}.$$

$$\text{From the second row, we get} \quad x_2 = (b_2^{(1)} - a_{23}^{(1)}x_3)/a_{22}^{(1)}.$$

$$\text{From the first row, we get} \quad x_1 = (b_1 - a_{12}x_2 - a_{13}x_3)/a_{11}.$$

In general, using a pivot, all the elements below that pivot in that column are made zeros.

Alternately, at each stage of elimination, we may also make the pivot as 1, by dividing that particular row by the pivot.

- R** When does the Gauss elimination method as described above fail? It fails when any one of the pivots is zero or it is a very small number, as the elimination progresses. If a pivot is zero, then division by it gives over flow error, since division by zero is not defined. If a pivot is a very small number, then division by it introduces large round-off errors and the solution may contain large errors.

For example, we may have the system

$$\begin{aligned} 2x_2 + 5x_3 &= 7 \\ 7x_1 + x_2 - 2x_3 &= 6 \\ 2x_1 + 3x_2 + 8x_3 &= 13 \end{aligned}$$

in which the first pivot is zero.

Pivoting Procedures How do we avoid computational errors in Gauss elimination? To avoid computational errors, we follow the procedure of *partial pivoting*. In the first stage of elimination, the first column of the augmented matrix is searched for the largest element in magnitude and brought as the first pivot by interchanging the first row of the augmented matrix (first equation) with the row (equation) having the largest element in magnitude. In the second stage of elimination, the second column is searched for the largest element in magnitude among the $n - 1$ elements leaving the first element, and this element is brought as the second pivot by interchanging the second row of the augmented matrix with the later row having the largest element in magnitude. This procedure is continued until the upper triangular system is obtained. Therefore, partial pivoting is done after every stage of elimination. There is another procedure called *complete pivoting*. In this procedure, we search the entire matrix $\mathbf{A}$ in the augmented matrix for the largest element in magnitude and bring it as the first pivot.

This requires not only an interchange of the rows, but also an interchange of the positions of the variables. It is possible that the position of a variable is changed a number of times during this pivoting. We need to keep track of the positions of all the variables. Hence, the procedure is computationally expensive and is not used in any software.

- R** Gauss elimination method is a direct method. Therefore, it is possible to count the total number of operations, that is, additions, subtractions, divisions and multiplications. Without going into details, we mention that the total number of divisions and multiplications (division and multiplication take the same amount of computer time) is $n(n^2 + 3n - 1)/3$. The total number of additions and subtractions (addition and subtraction take the same amount of computer time) is $n(n - 1)(2n + 5)/6$.
- R** When the system of algebraic equations is large, how do we conclude that it is consistent or not, using the Gauss elimination method? A way of determining the consistency is from the form of the reduced system (1.41). We know that if the system is inconsistent then $\text{rank}(\mathbf{A}) \neq \text{rank}[\mathbf{A} \mid \mathbf{b}]$. By checking the elements of the last rows, conclusion can be drawn about the consistency or inconsistency.

Suppose that in (1.41), $a_{33}^{(2)} \neq 0$ and $b_3^{(2)} \neq 0$. Then, $\text{rank}(\mathbf{A}) = \text{rank}[\mathbf{A} \mid \mathbf{b}] = 3$. The system is consistent and has a unique solution.

Suppose that we obtain the reduced system as

$$\left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ 0 & a_{22}^{(1)} & a_{23}^{(1)} & b_2^{(1)} \\ 0 & 0 & 0 & b_3^{(2)} \end{array} \right]$$

Then, $\text{rank}(\mathbf{A}) = 2, \text{rank}[\mathbf{A} \mid \mathbf{b}] = 3$ and $\text{rank}(\mathbf{A}) \neq \text{rank}[\mathbf{A} \mid \mathbf{b}]$. Therefore, the system is inconsistent and has no solution.

Suppose that we obtain the reduced system as

$$\left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ 0 & a_{22}^{(1)} & a_{23}^{(1)} & b_2^{(1)} \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Then, $\text{rank}(\mathbf{A}) = \text{rank}[\mathbf{A} \mid \mathbf{b}] = 2 < 3$. Therefore, the system has $3 - 2 = 1$ parameter family of infinite number of solutions.

2.3.1 Solved Problems

Problem 2.1 Solve the system of equations

$$3x_1 + 3x_2 + 4x_3 = 20$$

$$2x_1 + x_2 + 3x_3 = 13$$

$$x_1 + x_2 + 3x_3 = 6$$

using the Gauss elimination method.

Solution. Let us solve this problem by making the pivots as 1. The augmented matrix is given by

$$\left[\begin{array}{ccc|c} 3 & 3 & 4 & 20 \\ 2 & 1 & 3 & 13 \\ 1 & 1 & 3 & 6 \end{array} \right]$$

We perform the following elementary row transformations and do the eliminations.

$$R_1/3 \Rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 4/3 & 20/3 \\ 2 & 1 & 3 & 13 \\ 1 & 1 & 3 & 6 \end{array} \right]$$

$$R_2 - 2R_1, R_3 - R_1 \Rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 4/3 & 20/3 \\ 0 & -1 & 1/3 & -1/3 \\ 0 & 0 & 5/3 & -2/3 \end{array} \right]$$

Back substitution gives the solution as

$$x_3 = -\left(\frac{2}{3}\right)\left(\frac{3}{5}\right) = -\frac{2}{5}, x_2 = \frac{1}{3} + \frac{x_3}{3} = \frac{1}{3} + \frac{1}{3}\left(-\frac{2}{5}\right) = \frac{1}{5},$$

$$x_1 = \frac{20}{3} - x_2 - \frac{4}{3}x_3 = \frac{20}{3} - \frac{1}{5} - \frac{4}{3}\left(-\frac{2}{5}\right) = \frac{35}{5} = 7.$$

■

Problem 2.2 Test the consistency of the following system of equations

$$x_1 + 10x_2 - x_3 = 3$$

$$2x_1 + 3x_2 + 20x_3 = 7$$

$$9x_1 + 22x_2 + 79x_3 = 45$$

using the Gauss elimination method.

Solution. We have the augmented matrix as

$$\left[\begin{array}{ccc|c} 1 & 10 & -1 & 3 \\ 2 & 3 & 20 & 7 \\ 9 & 22 & 79 & 45 \end{array} \right]$$

We perform the following elementary row transformations and do the eliminations.

$$\begin{aligned} R_2 - 2R_1, R_3 - 9R_1 &\Rightarrow \left[\begin{array}{ccc|c} 1 & 10 & -1 & 3 \\ 0 & -17 & 22 & 1 \\ 0 & -68 & 88 & 18 \end{array} \right] \\ R_3 - 4R_2 &\Rightarrow \left[\begin{array}{ccc|c} 1 & 10 & -1 & 3 \\ 0 & -17 & 22 & 1 \\ 0 & 0 & 0 & 14 \end{array} \right] \end{aligned}$$

Now, $\text{rank}[\mathbf{A}] = 2$, and $\text{rank}[\mathbf{A} | \mathbf{b}] = 3$. Therefore, the system is inconsistent and has no solution. ■

2.4 Gauss-Jacobi Iteration Method

Sometimes, the method is called Jacobi method. We assume that the pivots $a_{ii} \neq 0$, for all i . Write the equations as

$$\begin{aligned} a_{11}x_1 &= b_1 - (a_{12}x_2 + a_{13}x_3) \\ a_{22}x_2 &= b_2 - (a_{21}x_1 + a_{23}x_3) \\ a_{33}x_3 &= b_3 - (a_{31}x_1 + a_{32}x_2) \end{aligned}$$

The Jacobi iteration method is defined as

$$\begin{aligned} x_1^{(k+1)} &= \frac{1}{a_{11}} \left[b_1 - (a_{12}x_2^{(k)} + a_{13}x_3^{(k)}) \right] \\ x_2^{(k+1)} &= \frac{1}{a_{22}} \left[b_2 - (a_{21}x_1^{(k)} + a_{23}x_3^{(k)}) \right] \\ x_3^{(k+1)} &= \frac{1}{a_{33}} \left[b_3 - (a_{31}x_1^{(k)} + a_{32}x_2^{(k)}) \right], \quad k = 0, 1, 2, \dots \end{aligned} \quad (2.10)$$

Since, we replace the complete vector $\mathbf{x}^{(k)}$ in the right hand side of (2.10) at the end of each iteration, this method is also called the method of simultaneous displacement.

R A sufficient condition for convergence of the Jacobi method is that the system of equations is diagonally dominant, that is, the coefficient matrix $\mathbf{A}$ is diagonally dominant. We can verify that $|a_{ii}| \geq \sum_{j=1, i \neq j}^n |a_{ij}|$. This implies that convergence may be obtained even if the system is not diagonally dominant. If the system is not diagonally dominant, we may exchange the equations, if possible, such that the new system is diagonally dominant and convergence is guaranteed. However, such manual verification or exchange of equations may not be possible for large systems that we obtain in application problems. The necessary and sufficient condition for convergence is that the spectral radius of the iteration matrix $\mathbf{H}$ is less than one unit, that is, $\rho(\mathbf{H}) < 1$, where $\rho(\mathbf{H})$ is the largest eigen value in magnitude of $\mathbf{H}$. Testing of this condition is beyond the scope of the syllabus.

R How do we find the initial approximations to start the iteration? If the system is diagonally dominant, then the iteration converges for any initial solution vector. If no suitable approximation is available, we can choose $\mathbf{x} = \mathbf{0}$, that is $x_i = 0$ for all i . Then, the initial approximation becomes $x_i = b_i/a_{ii}$, for all i .

2.4.1 Solved Problems**Problem 2.3** Solve the system of equations

$$\begin{aligned}4x_1 + x_2 + x_3 &= 2 \\x_1 + 5x_2 + 2x_3 &= -6 \\x_1 + 2x_2 + 3x_3 &= -4\end{aligned}$$

using the Jacobi iteration method. Use the initial approximations as

(i) $x_1 = 0.5, x_2 = -0.5, x_3 = -0.5$.(ii) $x_i = 0, i = 1, 2, 3$,

Perform five iterations in each case.

Solution. (i) $x_1^{(0)} = 0.5, x_2^{(0)} = -0.5, x_3^{(0)} = -0.5$.**First iteration**

$$\begin{aligned}x_1^{(1)} &= 0.25 \left[2 - (x_2^{(0)} + x_3^{(0)}) \right] = 0.25[2 - (-0.5 - 0.5)] = 0.75, \\x_2^{(1)} &= 0.2 \left[-6 - (x_1^{(0)} + 2x_3^{(0)}) \right] = 0.2[-6 - (0.5 + 2(-0.5))] = -1.1, \\x_3^{(1)} &= 0.33333 \left[-4 - (x_1^{(0)} + 2x_2^{(0)}) \right] = 0.33333[-4 - (0.5 + 2(-0.5))] = -1.16667.\end{aligned}$$

Second iteration

$$\begin{aligned}x_1^{(2)} &= 0.25 \left[2 - (x_2^{(1)} + x_3^{(1)}) \right] = 0.25[2 - (-1.1 - 1.16667)] = 1.06667, \\x_2^{(2)} &= 0.2 \left[-6 - (x_1^{(1)} + 2x_3^{(1)}) \right] = 0.2[-6 - (0.75 + 2(-1.16667))] = -0.88333, \\x_3^{(2)} &= 0.33333 \left[-4 - (x_1^{(1)} + 2x_2^{(1)}) \right] = 0.33333[-4 - (0.75 + 2(-1.1))] = -0.84999.\end{aligned}$$

Third iteration

$$\begin{aligned}x_1^{(3)} &= 0.25 \left[2 - (x_2^{(2)} + x_3^{(2)}) \right] = 0.25[2 - (-0.88333 - 0.84999)] = 0.93333, \\x_2^{(3)} &= 0.2 \left[-6 - (x_1^{(2)} + 2x_3^{(2)}) \right] = 0.2[-6 - (1.06667 + 2(-0.84999))] = -1.07334, \\x_3^{(3)} &= 0.33333 \left[-4 - (x_1^{(2)} + 2x_2^{(2)}) \right] \\&= 0.33333[-4 - (1.06667 + 2(-0.88333))] = -1.09999.\end{aligned}$$

Fourth iteration

$$\begin{aligned}x_1^{(4)} &= 0.25 \left[2 - (x_2^{(3)} + x_3^{(3)}) \right] = 0.25[2 - (-1.07334 - 1.09999)] = 1.04333, \\x_2^{(4)} &= 0.2 \left[-6 - (x_1^{(3)} + 2x_3^{(3)}) \right] = 0.2[-6 - (0.93333 + 2(-1.09999))] = -0.94667, \\x_3^{(4)} &= 0.33333 \left[-4 - (x_1^{(3)} + 2x_2^{(3)}) \right] \\&= 0.33333[-4 - (0.93333 + 2(-1.07334))] = -0.92887.\end{aligned}$$

Fifth iteration

$$\begin{aligned}x_1^{(5)} &= 0.25 \left[2 - (x_2^{(4)} + x_3^{(4)}) \right] = 0.25[2 - (-0.94667 - 0.92887)] = 0.96889, \\x_2^{(5)} &= 0.2 \left[-6 - (x_1^{(4)} + 2x_3^{(4)}) \right] = 0.2[-6 - (1.04333 + 2(-0.92887))] = -1.03712, \\x_3^{(5)} &= 0.33333 \left[-4 - (x_1^{(4)} + 2x_2^{(4)}) \right] \\&= 0.33333[-4 - (1.04333 + 2(-0.94667))] = -1.04999.\end{aligned}$$

(ii) Note that the given system is diagonally dominant. Jacobi method gives the iterations as

$$\begin{aligned}x_1^{(k+1)} &= 0.25 \left[2 - \left(x_2^{(k)} + x_3^{(k)} \right) \right] \\x_2^{(k+1)} &= 0.2 \left[-6 - \left(x_1^{(k)} + 2x_3^{(k)} \right) \right] \\x_3^{(k+1)} &= 0.33333 \left[-4 - \left(x_1^{(k)} + 2x_2^{(k)} \right) \right], \quad k = 0, 1, \dots\end{aligned}$$

We have the following results.

$$x_1^{(0)} = 0, x_2^{(0)} = 0, x_3^{(0)} = 0.$$

First iteration

$$\begin{aligned}x_1^{(1)} &= 0.25 \left[2 - \left(x_2^{(0)} + x_3^{(0)} \right) \right] = 0.5, \\x_2^{(1)} &= 0.2 \left[-6 - \left(x_1^{(0)} + 2x_3^{(0)} \right) \right] = -1.2, \\x_3^{(1)} &= 0.33333 \left[-4 - \left(x_1^{(0)} + 2x_2^{(0)} \right) \right] = -1.33333.\end{aligned}$$

Second iteration

$$\begin{aligned}x_1^{(2)} &= 0.25 \left[2 - \left(x_2^{(1)} + x_3^{(1)} \right) \right] = 0.25[2 - (-1.2 - 1.33333)] = 1.13333, \\x_2^{(2)} &= 0.2 \left[-6 - \left(x_1^{(1)} + 2x_3^{(1)} \right) \right] = 0.2[-6 - (0.5 + 2(-1.33333))] = -0.76668, \\x_3^{(2)} &= 0.33333 \left[-4 - \left(x_1^{(1)} + 2x_2^{(1)} \right) \right] = 0.33333[-4 - (0.5 + 2(-1.2))] = -0.7.\end{aligned}$$

Third iteration

$$\begin{aligned}x_1^{(3)} &= 0.25 \left[2 - \left(x_2^{(2)} + x_3^{(2)} \right) \right] = 0.25[2 - (-0.76668 - 0.7)] = 0.86667, \\x_2^{(3)} &= 0.2 \left[-6 - \left(x_1^{(2)} + 2x_3^{(2)} \right) \right] = 0.2[-6 - (1.13333 + 2(-0.7))] = -1.14667, \\x_3^{(3)} &= 0.33333 \left[-4 - \left(x_1^{(2)} + 2x_2^{(2)} \right) \right] \\&= 0.33333[-4 - (1.13333 + 2(-0.76668))] = -1.19998.\end{aligned}$$

Fourth iteration

$$\begin{aligned}x_1^{(4)} &= 0.25 \left[2 - \left(x_2^{(3)} + x_3^{(3)} \right) \right] = 0.25[2 - (-1.14667 - 1.19999)] = 1.08666, \\x_2^{(4)} &= 0.2 \left[-6 - \left(x_1^{(3)} + 2x_3^{(3)} \right) \right] = 0.2[-6 - (0.86667 + 2(-1.19998))] = -0.89334, \\x_3^{(4)} &= 0.33333 \left[-4 - \left(x_1^{(3)} + 2x_2^{(3)} \right) \right] \\&= 0.33333[-4 - (0.86667 + 2(-1.14667))] = -0.85777.\end{aligned}$$

Fifth iteration

$$\begin{aligned}x_1^{(5)} &= 0.25 \left[2 - \left(x_2^{(4)} + x_3^{(4)} \right) \right] = 0.25[2 - (-0.89334 - 0.85777)] = 0.93778, \\x_2^{(5)} &= 0.2 \left[-6 - \left(x_1^{(4)} + 2x_3^{(4)} \right) \right] = 0.2[-6 - (1.08666 + 2(-0.85777))] = -1.07422, \\x_3^{(5)} &= 0.33333 \left[-4 - \left(x_1^{(4)} + 2x_2^{(4)} \right) \right] \\&= 0.33333[-4 - (1.08666 + 2(-0.89334))] = -1.09998.\end{aligned}$$

It is interesting to note that the iterations oscillate and converge to the exact solution $x_1 = 1.0, x_2 = -1, x_3 = -1.0$. ■

Problem 2.4 Solve the system of equations

$$26x_1 + 2x_2 + 2x_3 = 12.6$$

$$3x_1 + 27x_2 + x_3 = -14.3$$

$$2x_1 + 3x_2 + 17x_3 = 6.0$$

using the Jacobi iteration method. Obtain the result correct to three decimal places.

Solution. The given system of equations is strongly diagonally dominant. Hence, we can expect faster convergence. Jacobi method gives the iterations as

$$\begin{aligned} x_1^{(k+1)} &= \left[12.6 - (2x_2^{(k)} + 2x_3^{(k)}) \right] / 26 \\ x_2^{(k+1)} &= \left[-14.3 - (3x_1^{(k)} + x_3^{(k)}) \right] / 27 \\ x_3^{(k+1)} &= \left[6.0 - (2x_1^{(k)} + 3x_2^{(k)}) \right] / 17 \quad k = 0, 1, \dots \end{aligned}$$

Choose the initial approximation as $x_1^{(0)} = 0, x_2^{(0)} = 0, x_3^{(0)} = 0$. We obtain the following results.

First iteration

$$x_1^{(1)} = \frac{1}{26} \left[12.6 - (2x_2^{(0)} + 2x_3^{(0)}) \right] = \frac{1}{26} [12.6] = 0.48462,$$

$$x_2^{(1)} = \frac{1}{27} \left[-14.3 - (3x_1^{(0)} + x_3^{(0)}) \right] = \frac{1}{27} [-14.3] = -0.52963$$

$$x_3^{(1)} = \frac{1}{17} \left[6.0 - (2x_1^{(0)} + 3x_2^{(0)}) \right] = \frac{1}{17} [6.0] = 0.35294.$$

Second iteration

$$x_1^{(2)} = \frac{1}{26} \left[12.6 - (2x_2^{(1)} + 2x_3^{(1)}) \right] = \frac{1}{26} [12.6 - 2(-0.52963 + 0.35294)] = 0.49821,$$

$$x_2^{(2)} = \frac{1}{27} \left[-14.3 - (3x_1^{(1)} + x_3^{(1)}) \right] = \frac{1}{27} [-14.3 - (3(0.48462) + 0.35294)] = -0.59655,$$

$$x_3^{(2)} = \frac{1}{17} \left[6.0 - (2x_1^{(1)} + 3x_2^{(1)}) \right] = \frac{1}{17} [6.0 - (2(0.48462) + 3(-0.52963))] = 0.38939.$$

Third iteration

$$x_1^{(3)} = \frac{1}{26} \left[12.6 - (2x_2^{(2)} + 2x_3^{(2)}) \right] = \frac{1}{26} [12.6 - 2(-0.59655 + 0.38939)] = 0.50006,$$

$$x_2^{(3)} = \frac{1}{27} \left[-14.3 - (3x_1^{(2)} + x_3^{(2)}) \right] = \frac{1}{27} [-14.3 - (3(0.49821) + 0.38939)] = -0.59941,$$

$$x_3^{(3)} = \frac{1}{17} \left[6.0 - (2x_1^{(2)} + 3x_2^{(2)}) \right] = \frac{1}{17} [6.0 - (2(0.49821) + 3(-0.59655))] = 0.39960.$$

Fourth iteration

$$x_1^{(4)} = \frac{1}{26} \left[12.6 - (2x_2^{(3)} + 2x_3^{(3)}) \right] = \frac{1}{26} [12.6 - 2(-0.59941 + 0.39960)] = 0.50000,$$

$$x_2^{(4)} = \frac{1}{27} \left[-14.3 - (3x_1^{(3)} + x_3^{(3)}) \right] = \frac{1}{27} [-14.3 - (3(0.50006) + 0.39960)] = -0.59999,$$

$$x_3^{(4)} = \frac{1}{17} \left[6.0 - (2x_1^{(3)} + 3x_2^{(3)}) \right] = \frac{1}{17} [6.0 - (2(0.50006) + 3(-0.59941))] = 0.39989.$$

We find

$$\begin{aligned} |x_1^{(4)} - x_1^{(3)}| &= |0.5 - 0.50006| = 0.00006, \\ |x_2^{(4)} - x_2^{(3)}| &= |-0.59999 + 0.59941| = 0.00058, \\ |x_3^{(4)} - x_3^{(3)}| &= |0.39989 - 0.39960| = 0.00029. \end{aligned}$$

Three decimal places of accuracy have not been obtained at this iteration.

Fifth iteration

$$\begin{aligned} x_1^{(5)} &= \frac{1}{26} [12.6 - (2x_2^{(4)} + 2x_3^{(4)})] = \frac{1}{26} [12.6 - 2(-0.59999 + 0.39989)] = 0.50001, \\ x_2^{(5)} &= \frac{1}{27} [-14.3 - (3x_1^{(4)} + x_3^{(4)})] = \frac{1}{27} [-14.3 - (3(0.50000) + 0.39989)] = -0.60000, \\ x_3^{(5)} &= \frac{1}{17} [-6.0 - (2x_1^{(4)} + 3x_2^{(4)})] = \frac{1}{17} [6.0 - (2(0.50000) + 3(-0.59999))] = 0.40000. \end{aligned}$$

We find

$$\begin{aligned} |x_1^{(4)} - x_1^{(3)}| &= |0.50001 - 0.5| = 0.00001, \\ |x_2^{(4)} - x_2^{(3)}| &= |-0.6 + 0.59999| = 0.00001, \\ |x_3^{(4)} - x_3^{(3)}| &= |0.4 - 0.39989| = 0.00011. \end{aligned}$$

Since, all the errors in magnitude are less than 0.0005, the required solution is

$$x_1 = 0.5, x_2 = -0.6, x_3 = 0.4.$$

■

R *What is the disadvantage of the Gauss-Jacobi method?* At any iteration step, the value of the first variable x_1 is obtained using the values of the previous iteration. The value of the second variable x_2 is also obtained using the values of the previous iteration, even though the updated value of x_1 is available. In general, at every stage in the iteration, values of the previous iteration are used even though the updated values of the previous variables are available. If we use the updated values of $x_1, x_2, \dots, x_{i-1}$ in computing the value of the variable x_i , then we obtain a new method called Gauss-Seidel iteration method.

2.5 Gauss-Seidel Iteration Method

As pointed out in Remark 22, we use the updated values of $x_1, x_2, \dots, x_{i-1}$ in computing the value of the variable x_i . We assume that the pivots $a_{ii} \neq 0$, for all i . We write the equations as

$$\begin{aligned} a_{11}x_1 &= b_1 - (a_{12}x_2 + a_{13}x_3) \\ a_{22}x_2 &= b_2 - (a_{21}x_1 + a_{23}x_3) \\ a_{33}x_3 &= b_3 - (a_{31}x_1 + a_{32}x_2) \end{aligned} \tag{2.11}$$

The Gauss-Seidel iteration method is defined as

$$\begin{aligned} x_1^{(k+1)} &= \frac{1}{a_{11}} [b_1 - (a_{12}x_2^{(k)} + a_{13}x_3^{(k)})] \\ x_2^{(k+1)} &= \frac{1}{a_{22}} [b_2 - (a_{21}x_1^{(k+1)} + a_{23}x_3^{(k)})] \\ x_3^{(k+1)} &= \frac{1}{a_{33}} [b_3 - (a_{31}x_1^{(k+1)} + a_{32}x_2^{(k+1)})], \quad k = 0, 1, 2, \dots \end{aligned} \tag{2.12}$$

This method is also called the method of successive displacement.

We observe that (2.11) is same as writing the given system as

$$\begin{aligned} a_{11}x_1^{(k+1)} &= b_1 - (a_{12}x_2^{(k)} + a_{13}x_3^{(k)}) \\ a_{21}x_1^{(k+1)} + a_{22}x_2^{(k+1)} &= b_2 - a_{23}x_3^{(k)} \\ a_{31}x_1^{(k+1)} + a_{32}x_2^{(k+1)} + a_{33}x_3^{(k+1)} &= b_3 \end{aligned}$$

R A sufficient condition for convergence of the Gauss-Seidel method is that the system of equations is diagonally dominant, that is, the coefficient matrix $\mathbf{A}$ is diagonally dominant. This implies that convergence may be obtained even if the system is not diagonally dominant. If the system is not diagonally dominant, we may exchange the equations, if possible, such that the new system is diagonally dominant and convergence is guaranteed. The necessary and sufficient condition for convergence is that the spectral radius of the iteration matrix $\mathbf{H}$ is less than one unit, that is, $\rho(\mathbf{H}) < 1$, where $\rho(\mathbf{H})$ is the largest eigen value in magnitude of $\mathbf{H}$. Testing of this condition is beyond the scope of the syllabus.

If both the Gauss-Jacobi and Gauss-Seidel methods converge, then Gauss-Seidel method converges at least two times faster than the Gauss-Jacobi method.

2.5.1 Solved Problems

Problem 2.5 Find the solution of the system of equations

$$\begin{aligned} 45x_1 + 2x_2 + 3x_3 &= 58 \\ -3x_1 + 22x_2 + 2x_3 &= 47 \\ 5x_1 + x_2 + 20x_3 &= 67 \end{aligned}$$

correct to three decimal places, using the Gauss-Seidel iteration method.

Solution. The given system of equations is strongly diagonally dominant. Hence, we can expect fast convergence. Gauss-Seidel method gives the iteration

$$\begin{aligned} x_1^{(k+1)} &= \frac{1}{45} (58 - 2x_2^{(k)} - 3x_3^{(k)}), \\ x_2^{(k+1)} &= \frac{1}{22} (47 + 3x_1^{(k+1)} - 2x_3^{(k)}), \\ x_3^{(k+1)} &= \frac{1}{20} (67 - 5x_1^{(k+1)} - x_2^{(k+1)}). \end{aligned}$$

Starting with $x_1^{(0)} = 0, x_2^{(0)} = 0, x_3^{(0)} = 0$, we get the following results.

First iteration

$$\begin{aligned} x_1^{(1)} &= \frac{1}{45} (58 - 2x_2^{(0)} - 3x_3^{(0)}) = \frac{1}{45} (58) = 1.28889, \\ x_2^{(1)} &= \frac{1}{22} (47 + 3x_1^{(1)} - 2x_3^{(0)}) = \frac{1}{22} (47 + 3(1.28889) - 2(0)) = 2.31212, \\ x_3^{(1)} &= \frac{1}{20} (67 - 5x_1^{(1)} - x_2^{(1)}) = \frac{1}{20} (67 - 5(1.28889) - (2.31212)) = 2.91217. \end{aligned}$$

Second iteration

$$\begin{aligned} x_1^{(2)} &= \frac{1}{45} (58 - 2x_2^{(1)} - 3x_3^{(1)}) = \frac{1}{45} (58 - 2(2.31212) - 3(2.91217)) = 0.99198, \\ x_2^{(2)} &= \frac{1}{22} (47 + 3x_1^{(2)} - 2x_3^{(1)}) = \frac{1}{22} (47 + 3(0.99198) - 2(2.91217)) = 2.00689, \end{aligned}$$

$$x_3^{(2)} = \frac{1}{20} (67 - 5x_1^{(2)}x_2^{(2)}) = \frac{1}{20} (67 - 5(0.99198) - (2.00689)) = 3.00166.$$

Third iteration

$$x_2^{(3)} = \frac{1}{22} (47 + 3x_1^{(3)}2x_3^{(2)}) = \frac{1}{22} (47 + 3(0.99958) - 2(3.00166)) = 1.99979,$$

$$x_3^{(3)} = \frac{1}{20} (67 - 5x_1^{(3)}x_2^{(3)}) = \frac{1}{20} (67 - 5(0.99958) - (1.99979)) = 3.00012.$$

Fourth iteration

$$x_1^{(4)} = \frac{1}{45} (58 - 2x_2^{(3)}3x_3^{(3)}) = \frac{1}{45} (58 - 2(1.99979) - 3(3.00012)) = 1.00000,$$

$$x_2^{(4)} = \frac{1}{22} (47 + 3x_1^{(4)}2x_3^{(3)}) = \frac{1}{22} (47 + 3(1.00000) - 2(3.00012)) = 1.99999,$$

$$x_3^{(4)} = \frac{1}{20} (67 - 5x_1^{(4)}x_2^{(4)}) = \frac{1}{20} (67 - 5(1.00000) - (1.99999)) = 3.00000.$$

We find

$$\begin{aligned} |x_1^{(4)} - x_1^{(3)}| &= |1.00000 - 0.99958| = 0.00042, \\ |x_2^{(4)} - x_2^{(3)}| &= |1.99999 - 1.99979| = 0.00020, \\ |x_3^{(4)} - x_3^{(3)}| &= |3.00000 - 3.00012| = 0.00012. \end{aligned}$$

Since, all the errors in magnitude are less than 0.0005, the required solution is

$$x_1 = 1.0, x_2 = 1.99999, x_3 = 3.0.$$

Rounding to three decimal places, we get $x_1 = 1.0, x_2 = 2.0, x_3 = 3.0$. ■

Problem 2.6 Computationally show that Gauss-Seidel method applied to the system of equations

$$\begin{aligned} 3x_1 - 6x_2 + 2x_3 &= 23 \\ -4x_1 + x_2 - x_3 &= -8 \\ x_1 - 3x_2 + 7x_3 &= 17 \end{aligned}$$

diverges. Take the initial approximations as $x_1 = 0.9, x_2 = -3.1, x_3 = 0.9$. Interchange the first and second equations and solve the resulting system by the Gauss-Seidel method. Again take the initial approximations as $x_1 = 0.9, x_2 = -3.1, x_3 = 0.9$, and obtain the result correct to two decimal places. The exact solution is $x_1 = 1.0, x_2 = -3.0, x_3 = 1.0$.

Solution. Note that the system of equations is not diagonally dominant. Gauss-Seidel method gives the iteration

$$\begin{aligned} x_1^{(k+1)} &= [23 + 6x_2^{(k)} - 2x_3^{(k)}] / 3 \\ x_2^{(k+1)} &= [-8 + 4x_1^{(k+1)} + x_3^{(k)}] \\ x_3^{(k+1)} &= [-17 - x_1^{(k+1)} + 3x_2^{(k+1)}] / 7 \end{aligned}$$

Starting with the initial approximations $x_1 = 0.9, x_2 = -3.1, x_3 = 0.9$, we obtain the following results.

First iteration

$$\begin{aligned}x_1^{(1)} &= \frac{1}{3} [23 + 6x_2^{(0)} - 2x_3^{(0)}] = \frac{1}{3} [23 + 6(-3.1) - 2(0.9)] = 0.8667, \\x_2^{(1)} &= [-8 + 4x_1^{(1)} + x_3^{(0)}] = [-8 + 4(0.8667) + 0.9] = -3.6332, \\x_3^{(1)} &= \frac{1}{7} [17 - x_1^{(1)} + 3x_2^{(1)}] = \frac{1}{7} [17 - (0.8667) + 3(-3.6332)] = 0.7477.\end{aligned}$$

Second iteration

$$\begin{aligned}x_1^{(2)} &= \frac{1}{3} [23 + 6x_2^{(1)} - 2x_3^{(1)}] = \frac{1}{3} [23 + 6(-3.6332) - 2(0.7477)] = -0.0982 \\x_2^{(2)} &= [-8 + 4x_1^{(2)} + x_3^{(1)}] = [-8 + 4(-0.0982) + 0.7477] = -7.6451 \\x_3^{(2)} &= \frac{1}{7} [17 - x_1^{(2)} + 3x_2^{(2)}] = \frac{1}{7} [17 + 0.0982 + 3(-7.6451)] = -0.8339\end{aligned}$$

Third iteration

$$\begin{aligned}x_1^{(3)} &= \frac{1}{3} [23 + 6x_2^{(2)} - 2x_3^{(2)}] = \frac{1}{3} [23 + 6(-7.6451) - 2(-0.8339)] = -7.0676 \\x_2^{(3)} &= [-8 + 4x_1^{(3)} + x_3^{(2)}] = [-8 + 4(-7.0676) - 0.8339] = -37.1043 \\x_3^{(3)} &= \frac{1}{7} [17 - x_1^{(3)} + 3x_2^{(3)}] = \frac{1}{7} [17 + 7.0676 + 3(-37.1043)] = -12.4636\end{aligned}$$

It can be observed that the iterations are diverging very fast.

Now, we exchange the first and second equations to obtain the system

$$\begin{aligned}-4x_1 + x_2 - x_3 &= -8 \\3x_1 - 6x_2 + 2x_3 &= 23 \\x_1 - 3x_2 + 7x_3 &= 17\end{aligned}$$

The system of equations is now diagonally dominant. Gauss-Seidel method gives iteration

$$\begin{aligned}x_1^{(k+1)} &= [8 + x_2^{(k)} - x_3^{(k)}] / 4 \\x_2^{(k+1)} &= -[23 - 3x_1^{(k+1)} + 2x_3^{(k)}] / 6 \\x_3^{(k+1)} &= [17 - x_1^{(k+1)} + 3x_2^{(k+1)}] / 7\end{aligned}$$

Starting with the initial approximations $x_1 = 0.9$, $x_2 = -3.1$, $x_3 = 0.9$, we obtain the following results. First iteration

$$\begin{aligned}x_1^{(1)} &= \frac{1}{4} [8 + x_2^{(0)} - x_3^{(0)}] = \frac{1}{4} [8 - 3.1 - 0.9] = 1.0, \\x_2^{(1)} &= -\frac{1}{6} [23 - 3x_1^{(1)} + 2x_3^{(0)}] = -\frac{1}{6} [23 - 3(1.0) - 2(0.9)] = -3.0333, \\x_3^{(1)} &= \frac{1}{7} [17 - x_1^{(1)} + 3x_2^{(1)}] = \frac{1}{7} [17 - 1.0 + 3(-3.0333)] = 0.9857.\end{aligned}$$

Second iteration

$$\begin{aligned}x_1^{(2)} &= \frac{1}{4} [8 + x_2^{(1)} - x_3^{(1)}] = \frac{1}{4} [8 - 3.0333 - 0.9857] = 0.9953, \\x_2^{(2)} &= -\frac{1}{6} [23 - 3x_1^{(2)} + 2x_3^{(1)}] = -\frac{1}{6} [23 - 3(0.9953) - 2(0.9857)] = -3.0071, \\x_3^{(2)} &= \frac{1}{7} [17 - x_1^{(2)} + 3x_2^{(2)}] = \frac{1}{7} [17 - 0.9953 + 3(-3.0071)] = 0.9976.\end{aligned}$$

Third iteration

$$\begin{aligned}x_1^{(3)} &= \frac{1}{4} \left[8 + x_2^{(2)} - x_3^{(2)} \right] = \frac{1}{4} [8 - 3.0071 - 0.9976] = 0.9988, \\x_2^{(3)} &= -\frac{1}{6} \left[23 - 3x_1^{(3)} - 2x_3^{(2)} \right] = -\frac{1}{6} [23 - 3(0.9988) - 2(0.9976)] = -3.0014, \\x_3^{(3)} &= \frac{1}{7} \left[17 - x_1^{(3)} + 3x_2^{(3)} \right] = \frac{1}{7} [17 - 0.9988 + 3(-3.0014)] = 0.9996.\end{aligned}$$

Fourth iteration

$$\begin{aligned}x_1^{(4)} &= \frac{1}{4} \left[8 + x_2^{(3)} - x_3^{(3)} \right] = \frac{1}{4} [8 - 3.0014 - 0.9996] = 0.9998, \\x_2^{(4)} &= -\frac{1}{6} \left[23 - 3x_1^{(4)} - 2x_3^{(3)} \right] = -\frac{1}{6} [23 - 3(0.9998) - 2(0.9996)] = -3.0002, \\x_3^{(4)} &= \frac{1}{7} \left[17 - x_1^{(4)} + 3x_2^{(4)} \right] = \frac{1}{7} [17 - 0.9998 + 3(-3.0002)] = 0.9999.\end{aligned}$$

We find

$$\begin{aligned}\left| x_1^{(4)} - x_1^{(3)} \right| &= |0.9998 - 0.9988| = 0.0010, \\ \left| x_2^{(4)} - x_2^{(3)} \right| &= |-3.0002 + 3.0014| = 0.0012, \\ \left| x_3^{(4)} - x_3^{(3)} \right| &= |0.9999 - 0.9996| = 0.0003,\end{aligned}$$

Since, all the errors in magnitude are less than 0.005, the required solution is

$$x_1 = 0.9998, x_2 = -3.0002, x_3 = 0.9999.$$

Rounding to two decimal places, we get $x_1 = 1.0, x_2 = -3.0, x_3 = 1.0$. ■